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Convolutional Neural Networks - II

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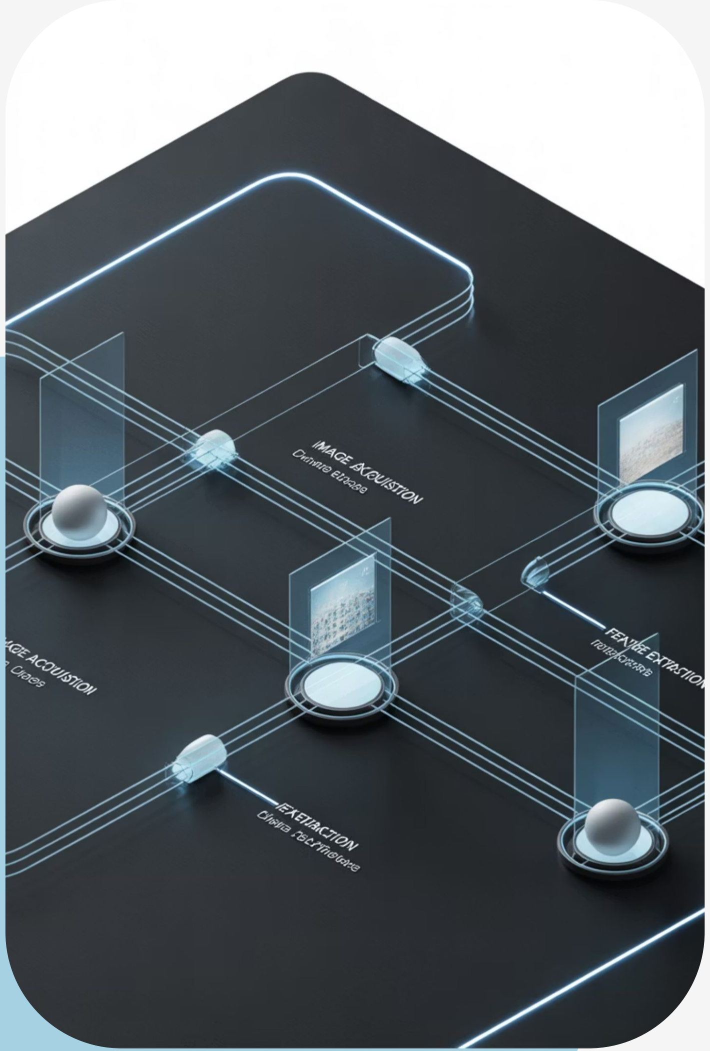
Session Overview

Key concepts in modern deep learning



Residual Connections

ResNet uses skip connections to solve vanishing gradient problems

Transfer Learning

Use pre-trained models for Dogs vs Cats with minimal training

Object Detection

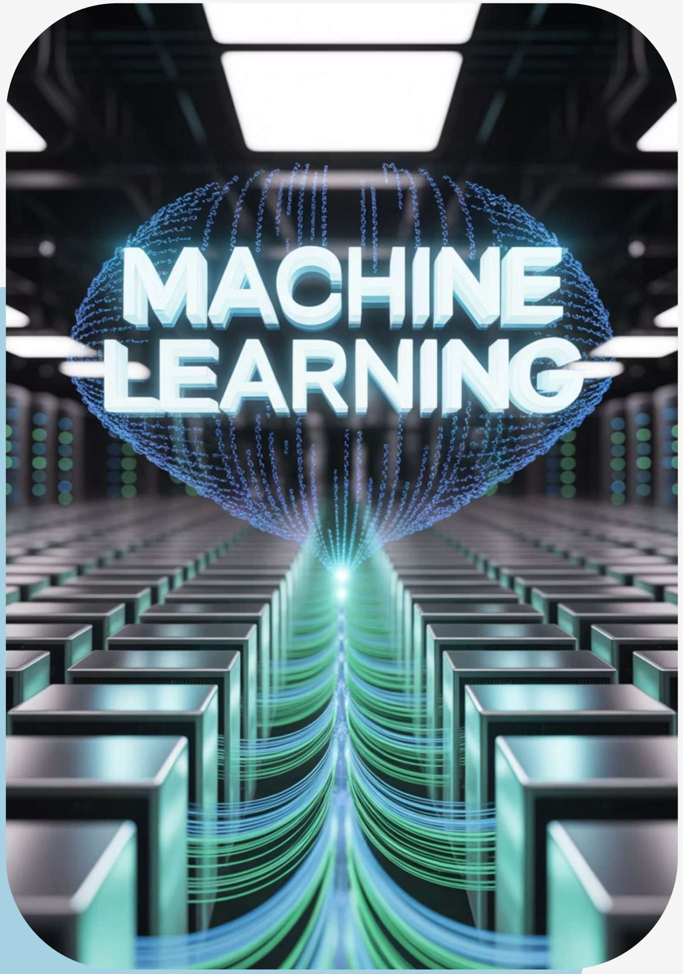
Detect objects in real time using YOLO

Image Segmentation

Classify pixels for precise object boundaries

Learning Objectives

By the end of this hands-on session, you will:



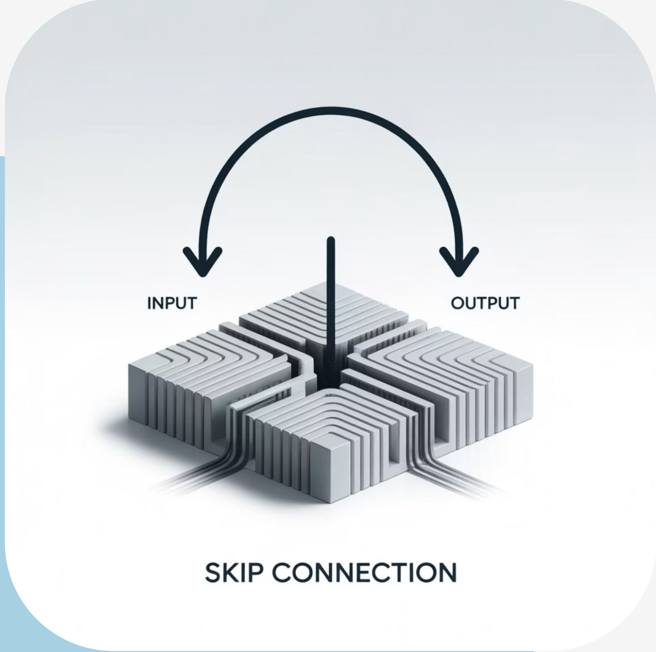
1. **ResNet Architecture**
   Residual connections, through skip connections, allow deeper networks to train effectively
2. **Transfer Learning**
   Apply VGG16 on the Dogs vs Cats dataset
3. **YOLO Detection**
   Use YOLO for real-time object detection with bounding boxes
4. **Image Segmentation**
   Perform pixel-level classification for scene understanding

Demo 1: Residual Connections (ResNet)

Overcoming The Vanishing Gradient Problem

The Vanishing Gradient Problem

Deep networks face vanishing gradients, limiting effective training



- **Problem**: Traditional deep networks lose performance as they get deeper
- **ResNet Solution**: Skip connections enable direct gradient flow
- **Results**: Faster convergence and higher accuracy on CIFAR-10

Demo: Train a simple CNN and ResNet on CIFAR-10 to compare validation accuracy, showing ResNet converges better

Demo 2: Transfer Learning (Dogs vs Cats)

1

Problem
Training CNNs from scratch needs large datasets and high compute

2

Solution
Use pre-trained VGG16 for feature extraction

3

Steps
Freeze base, add dense layers for binary classification

4

Training
Train only top layers, keep base frozen

Insight: Higher accuracy with less compute and training time

Demo 3: Object Detection with YOLO

Real-time detection in a single step

YOLO (You Only Look Once) revolutionizes object detection by treating it as a single regression problem

Architecture: YOLOv8 pre-trained on COCO with 80 classes

Predictions: Outputs classes and bounding box coordinates

Demo: Inference on test images (bus, Zidane) with original vs detected

Demo 4: Image Segmentation

Pixel-level understanding of images

Pixel Classification: Labels every pixel for precise boundaries

YOLOv8-seg: Combines detection with mask generation

Scene Insight: Captures fine detail and overlapping regions

The bus demo shows pixel-perfect boundaries, highlighting the model's detailed scene understanding.

Key Takeaways

Core insights from the session

ResNet: Skip connections overcome vanishing gradients for deeper networks

Transfer Learning: Pre-trained models speed up training and boost accuracy

YOLO: Real-time detection with class and location in one pass

Segmentation: Pixel-level classification for detailed scene understanding

Q&A Session

- Open floor for questions and clarifications.